

The empirical recurrence for OEIS A297737: recovering a conjecture that is not written down

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Abstract

OEIS A297737 records “Empirical recurrence of order 80”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 118$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 80, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 80 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297737 is “Number of $5 \times n$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 2 or 4 neighboring 1s.”. It has offset 1 and begins

1, 6, 151, 502, 2719, 11341, 82549, 396787, ...

The entry states:

Empirical recurrence of order 80 (see link above)

As of the “Last modified” line on the live entry (Jan 04 2018 22:21:15, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 118$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 118$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 236$ exact terms of the model returns order 80 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{80} c_i a(n-i),$$

c_1	11	c_{21}	17647702139	c_{41}	3375287869496	c_{61}	−3177094860824
c_2	−3	c_{22}	−49185963758	c_{42}	57298374456009	c_{62}	2291968920020
c_3	−341	c_{23}	36879992920	c_{43}	−54284751772471	c_{63}	2052162564504
c_4	1136	c_{24}	19205187580	c_{44}	98091899623604	c_{64}	−2623677858320
c_5	716	c_{25}	−195256268347	c_{45}	−30381744122443	c_{65}	3271112203432
c_6	−13648	c_{26}	527465399298	c_{46}	−40380942576546	c_{66}	−1787927107560
c_7	62387	c_{27}	−393049900573	c_{47}	45200301515035	c_{67}	702379047088
c_8	−95194	c_{28}	−25425460756	c_{48}	−128607836130125	c_{68}	93081657888
c_9	−422133	c_{29}	1288561438147	c_{49}	46051204945968	c_{69}	−340967776112
c_{10}	1840696	c_{30}	−3600643901415	c_{50}	−11379102131434	c_{70}	119095217504
c_{11}	−3381535	c_{31}	2796844101581	c_{51}	−10741397415239	c_{71}	−199951698624
c_{12}	2069563	c_{32}	−1230694904109	c_{52}	91416746145326	c_{72}	72894655744
c_{13}	28760185	c_{33}	−4822040660830	c_{53}	−21975815556938	c_{73}	−64426946432
c_{14}	−99809651	c_{34}	15476898801228	c_{54}	39674435224834	c_{74}	19822418368
c_{15}	108369995	c_{35}	−12862851542301	c_{55}	290757058696	c_{75}	5496426880
c_{16}	29830958	c_{36}	11503483150174	c_{56}	−24346167566362	c_{76}	−9946430848
c_{17}	−945858643	c_{37}	7877001071582	c_{57}	2916698332216	c_{77}	4830979072
c_{18}	2871514502	c_{38}	−39714176234206	c_{58}	−18497662483834	c_{78}	−825738240
c_{19}	−2380809639	c_{39}	35281708556995	c_{59}	−3984496255144	c_{79}	337408000
c_{20}	−1627303125	c_{40}	−45777452254049	c_{60}	1396752641992	c_{80}	259368960

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 80, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $80 + 4$ terms removed returns order 80 as well, so $\deg q_{\min} = 80 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 21 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $80 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 80 that this sequence satisfies — and that family turns out to have exactly one member.

References

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